Lab Report – EE-170L Computer Programming (Lab 04)

NAME	saifullah.
ID	725
SEMESTER	2nd
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Objective

The main objectives of this lab are:

• To learn how the switch statement works in C++.

• To understand the use of the break statement in controlling program execution.

• To study how the continue statement works inside loops.

• To identify the difference between break and continue statements.

• To create simple menu-based programs using switch statements.

Theory

Switch Statement

• The switch statement allows a program to choose one block of code from several options depending on the value of a variable.

• It is commonly used as a replacement for multiple if-else statements when checking a single variable against different values.

• Each case represents a possible value of the expression. When the value matches a case, its code is executed.

- The default case runs when none of the cases match the given value.

Syntax:

```
switch (expression) {
```

```
  case value1:
    // code
    break;
  case value2:
    // code
    break;
  default:
    // code
}
```

Break Statement

- Ends the execution of the current case.
- Prevents fall-through into subsequent cases.
- Without, execution continues until the next break or the end of the switch block.

Continue Statement

- Used inside loops.
- Skips the current iteration and moves to the next iteration of the loop.

Programs and Implementation

Program 1: Day of the Week

```
#include <iostream>
using namespace std;

int main() {
   int day;
   cout << "Enter day number (1-7): ";
   cin >> day;

   switch(day) {
      case 1: cout << "Monday"; break;
      case 2: cout << "Tuesday"; break;
      case 3: cout << "Wednesday"; break;
      case 4: cout << "Thursday"; break;
      case 5: cout << "Friday"; break;
      case 6: cout << "Saturday"; break;
      case 7: cout << "Sunday"; break;
      default: cout << "Invalid day!";
   }
   return 0;
}
```

Sample output:

Enter day number (1-7): 3 Wednesday

Program 2: Grade Evaluation

```
#include <iostream>
using namespace std;

int main () {
   char grade;
   cout << "Enter grade (A, B, C, D, F): ";
   cin >> grade;

   switch(grade) {
      case 'A': cout << "Excellent"; break;
      case 'B': cout << "Very Good"; break;
      case 'C': cout << "Good"; break;
      case 'D': cout << "Pass"; break;
      case 'F': cout << "Fail"; break;
      default: cout << "Invalid Grade!";
   }
   return 0;
}
```

Sample output:

Enter grade (A, B, C, D, F): B very good

Program 3: Conversion Menu

```
#include <iostream>
using namespace std;

int main() {
    int choice;
    double value;

    cout << "Menu:\n";
    cout << "1 -> Kilometers to Meters\n";
    cout << "2 -> Meters to Centimeters\n";
    cout << "3 -> Kilograms to Grams\n";
    cout << "4 -> Celsius to Fahrenheit\n";
    cout << "Enter your choice: ";
    cin >> choice;

    switch(choice) {
        case 1:
            cout << "Enter kilometers: ";
            cin >> value;
            cout << "Meters = " << value * 1000;
```

```
        break;
    case 2:
        cout << "Enter meters: ";
        cin >> value;
        cout << "Centimeters = " << value * 100;
        break;
    case 3:
        cout << "Enter kilograms: ";
        cin >> value;
        cout << "Grams = " << value * 1000;
        break;
    case 4:
        cout << "Enter Celsius: ";
        cin >> value;
        cout << "Fahrenheit = " << (value * 9/5) + 32;
        break;
    default:
        cout << "Invalid choice!";
  }
  return 0;
}
```

Output sample:

Menu:

1 -> Kilometers to Meters

2 -> Meters to Centimeters

3 -> Kilograms to Grams

4 -> Celsius to Fahrenheit

Enter your choice: 4

Enter Celsius: 25

Fahrenheit = 77

Results

The switch statement makes it easier to choose between multiple options.

• The break statement ensures that only the required case is executed.

• The default case is useful for handling incorrect or unexpected input values.

• Menu-driven applications can be easily created using switch statements.

Conclusion

We learned the concept and working of the switch–case statement in C++ for selecting one option from multiple choices.

We understood the role of the break statement in stopping execution and controlling the flow of the program.

We studied how the default case is used to handle invalid or unmatched inputs.

By creating programs like day selection, grade evaluation, and unit conversion, we understood the practical use of switch statements in programming.